Mahmoud Alyosify

AI Research Engineer

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SUMMARY

AI research engineer who runs **careful experiments**: a clear question, fair baselines, ablations and results anyone can reproduce. MSc in Artificial Intelligence, Queen's University, Canada (GPA 4.30 / 4.30); my thesis, MinimalLM, made three open LLMs generate 47.6% fewer tokens with no answer cut off. Also studies coding agents and self-supervised vision.

SKILLS

Research: experiment design, ablation studies, statistical testing, LLM evaluation (MT-Bench, AlpacaEval, LLM-as-judge)
Areas: efficient LLM inference, self-supervised learning, reinforcement learning
Tools: PyTorch, Hugging Face Transformers, scikit-learn, Gymnasium
Engineering: Python, Git, Linux, NVIDIA GPUs, AWS

RESEARCH PROJECTS

MinimalLM — Master's thesis (Python, PyTorch, Hugging Face Transformers) 2026

- Question: can an unchanged LLM write fewer tokens and still give a complete answer? Tested on 330 questions (MT-Bench, AlpacaEval, Arena-Hard-Auto) and six other languages.
- A training-free method that makes open LLMs (Mistral-7B, Llama-3.1-8B, Qwen2.5-3B) give shorter answers that are still complete: **47.6% fewer tokens, 19–68% faster, no answer cut off** (0 of 990).
- Honest limits: shorter answers more often leave out part of a broad question (22.9% vs 12.6% by the judge), and the evaluation is internal, with no independent human study.

Human vs. AI-Agent Pull Requests — research study (Python, GitHub data mining)

- Built a reproducible pipeline that mines GitHub to compare review effort on pull requests written by AI agents and by people.

SimCLR-Vision-SSL — self-supervised image model (PyTorch, ONNX, FAISS) 2026

- Ran a controlled 42-experiment study to find which image augmentations actually improve the model; best result 84.30% top-1 accuracy.

EXPERIENCE

AI Engineer Intern — Injaz Tech (Injaz Group), Cairo Oct 2026 – Present

- Designed the evaluation: retrieval benchmarks (top-1, MRR), category-level scores and an LLM judge checked against a fixed metric (agreement 0.82).
- Raised top-1 retrieval accuracy from **75% to 94%** and cut false refusals from 5 in 20 to 1 in 20.

Part-Time Machine Learning Instructor — National Telecommunication Institute (NTI), Egypt Jul 2025 – Present

- Teach **90-hour machine learning bootcamps** and AI for Business modules in NTI's Summer Internships Program.

Online Instructor — Udemy & YouTube (Arabic courses), Remote 2020 – Present

- Created Arabic courses on algorithms, discrete mathematics, probability and statistics: **10,000+ learners, 700K+ views**.

EDUCATION

MSc in Artificial Intelligence — Queen's University, Kingston, Canada Sep 2025 – Oct 2026

- GPA **4.30 / 4.30** · Completed October 2026 · Digilians Presidential Scholarship
- Thesis: MinimalLM — shorter LLM answers that stay complete (supervisor: Dr. Rahatara Ferdousi)

BSc in Computer & Information Science (Bioinformatics) — Assiut University, Egypt Sep 2019 – Jul 2023

- GPA 3.53 / 4.00 (Very Good with Honours) · Graduation project graded A+

CERTIFICATIONS & AWARDS

Deep Learning Specialization (DeepLearning.AI, 2026) · Machine Learning Specialization (DeepLearning.AI) · 1st Place, Smart Cities Hackathon 2022 (77 university teams) · Semi-Finalist, Microsoft Imagine Cup 2023

LANGUAGES

Arabic (native) · English (professional working proficiency)